

- Splunk UBA 5.4.2 Troubleshooting Guide
 - General Troubleshooting Commands
 - Basic System Status Check
 - Log File Locations
 - Installation Issues
 - “setup containerization failed” Error
 - Permission Issues During Installation
 - RHEL 8.10 Specific Issues
 - Service Issues
 - Service Won’t Start
 - Port Conflicts
 - Memory or CPU Issues
 - Data Source Issues
 - No Data Flowing
 - Authentication Failures
 - Identity Resolution Issues
 - SSL Certificate Issues
 - Browser Certificate Errors
 - Service Communication Certificate Issues
 - Performance Issues
 - High CPU/Memory Usage
 - Disk Space Issues
 - Slow Web Interface
 - UBA Monitor API Errors
 - Network Issues
 - Service Communication Issues
 - RHEL 8.10 Network Configuration Issues
 - Emergency Recovery Procedures
 - Reset Admin Password
 - Complete Service Reset
 - System Recovery After Reboot
 - Getting Help
 - Collecting Diagnostic Information
 - Support Information
 - References
 - UBA Monitor API Errors Section

Splunk UBA 5.4.2 Troubleshooting Guide

This guide provides comprehensive troubleshooting procedures for common issues encountered during and after Splunk UBA 5.4.2 deployment on RHEL 8.10.

General Troubleshooting Commands

Basic System Status Check

```
# Check UBA service status
/opt/caspida/bin/Caspida status

# Run comprehensive health check
/opt/caspida/bin/utils/uba_health_check.sh

# Check system resources
top
free -m
df -h
```

Log File Locations

```
# UI logs
tail -f /var/vcap/sys/log/caspida-ui/caspida-ui.stderr.log

# Job manager logs
tail -f /var/vcap/sys/log/caspida-jobmanager/caspida-jobmanager.stderr.log

# Data source logs
tail -f /var/vcap/sys/log/caspida-datasource/caspida-datasource.stderr.log

# System logs
tail -f /var/log/caspida/caspida.out

# Kafka logs
tail -f /var/vcap/sys/log/kafka/server.log

# Zookeeper logs
tail -f /var/vcap/sys/log/zookeeper/zookeeper.out
```

Installation Issues

“setup containerization failed” Error

If you encounter this error during installation or upgrade:

1. Check logs:

```
cat /var/log/caspida/caspida.out
```

2. Common causes and solutions:

Disk space issue:

```
# Check disk space
df -h

# Free up space if needed
rm -rf /tmp/*
sudo yum clean all
```

HTTP proxy configuration issue:

```
# Check if proxy is configured
env | grep -i proxy

# Unset proxy if causing issues
unset http_proxy https_proxy HTTP_PROXY HTTPS_PROXY
```

Docker service issue:

```
# Check Docker status
sudo systemctl status docker

# Restart Docker if needed
sudo systemctl restart docker
```

3. Fix and re-run:

```
# After fixing issues
/opt/caspida/bin/Caspida setup-containerization
```

Permission Issues During Installation

If installation fails due to permission errors:

1. Verify caspida user permissions:

```
# Check user and group
id caspida

# Verify sudo configuration
sudo -u caspida sudo -l
```

2. Fix ownership issues:

```
sudo chown -R caspida:caspida /opt/caspida
sudo chown -R caspida:caspida /home/caspida
```

3. Verify mount point permissions:

```
ls -la /var/vcap /var/vcap2
sudo chmod 755 /var/vcap /var/vcap2
```

RHEL 8.10 Specific Issues

Bridge Netfilter Module Not Loading

```
# Load the module
sudo modprobe br_netfilter

# Make it persistent
echo br_netfilter | sudo tee /etc/modules-load.d/br_net_filter.conf

# Configure bridge settings
sudo sysctl -w net.bridge.bridge-nf-call-iptables=1
echo net.bridge.bridge-nf-call-iptables=1 | sudo tee /etc/sysctl.d/splunkuba-bridge.conf
```

Firewall Configuration Issues

```
# Check firewall status
sudo firewall-cmd --state

# Verify interface zone assignment
sudo firewall-cmd --get-zone-of-interface=<interface_name>

# Assign interface to public zone if needed
sudo firewall-cmd --zone=public --add-interface=<interface_name> --permanent
sudo firewall-cmd --reload
```

Service Issues

Service Won’t Start

If specific services fail to start:

- 1. Check individual service status:

```
sudo systemctl status caspida-ui
sudo systemctl status kafka-server
sudo systemctl status postgresql
```

- 2. Check service logs:

```
# UI service
journalctl -u caspida-ui -f

# Kafka service
journalctl -u kafka-server -f
```

- 3. Restart problematic services:

```
sudo systemctl restart caspida-ui
sudo systemctl restart kafka-server
```

Port Conflicts

If services fail due to port conflicts:

- 1. Check for port conflicts:

```
netstat -tulpn | grep LISTEN
ss -tulpn | grep LISTEN
```

- 2. Identify conflicting processes:

```
sudo lsof -i :9002 # Check Job Manager port
sudo lsof -i :9092 # Check Kafka port
```

- 3. Stop conflicting services:

```
sudo systemctl stop <conflicting_service>
sudo systemctl disable <conflicting_service>
```

Memory or CPU Issues

If services fail due to resource constraints:

- 1. Check system resources:

```
top
htop
free -m
```

2. Check for memory-intensive processes:

```
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | head -20
```

3. Restart resource monitor:

```
sudo systemctl restart caspida-resourcesmonitor
```

Data Source Issues

No Data Flowing

If no data appears to be flowing from data sources:

1. Verify data source status in UI:

```
# Navigate to Data Sources view in UBA UI
# Check EPS (Events Per Second) - should be non-zero
```

2. Check data source logs:

```
grep "events processed" /var/vcap/sys/log/caspida-datasource/*.log
grep ERROR /var/vcap/sys/log/caspida-datasource/*.log
```

3. Test connectivity to Splunk:

```
# For Splunk data sources
curl -k https://splunk-hostname:8089
telnet splunk-hostname 8089
```

4. Verify SSL configuration (if applicable):

```
# Check if certificates are properly imported
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo keytool -list -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit | grep -i splunk
```

Authentication Failures

If data sources fail due to authentication:

1. Test Splunk credentials:

```
curl -k -u username:password https://splunk-hostname:8089/services/auth/login
```

2. Check for expired tokens:

```
grep "authentication" /var/vcap/sys/log/caspida-datasource/*.log
```

3. Verify user permissions in Splunk:

- Ensure user has appropriate roles (admin, power, uba_user)
- Check index access permissions

Identity Resolution Issues

If users are not properly associated with devices:

1. Check IDR exclusion list:

```
# Navigate to Manage > IDR Exclusions in UBA UI
```

2. Verify HR data is properly loaded:

```
# Navigate to Manage > HR Data in UBA UI
```

3. Adjust IDR thresholds if necessary:

```
# Edit /etc/caspida/local/conf/uba-site.properties
identity.resolution.blacklist.threshold.device.hostnamecount=2
identity.resolution.blacklist.threshold.device.hostnamehours=6

# Synchronize changes
/opt/caspida/bin/Caspida sync-cluster /etc/caspida/local/conf
```

SSL Certificate Issues

Browser Certificate Errors

If browser shows SSL certificate errors:

- 1. **Verify certificate configuration:**

```
# Check certificate properties in uba-site.properties
grep ui.auth /etc/caspida/local/conf/uba-site.properties
```

- 2. **Validate certificate file permissions:**

```
ls -la /var/vcap/store/caspida/certs/
```

- 3. **Check certificate validity:**

```
openssl x509 -in /var/vcap/store/caspida/certs/my-server.crt.pem -text -noout | grep -E 'Not Before|Not After'
```

- 4. **Regenerate self-signed certificate if expired:**

```
rm /var/vcap/store/caspida/certs/my-root-ca.crt.pem
/opt/caspida/bin/CaspidaCert.sh US CA "San Francisco" Splunk "" "" /var/vcap/store/caspida/certs/
sudo systemctl restart caspida-ui
```

Service Communication Certificate Issues

If services have SSL communication errors:

- 1. **Check Java keystore:**

```
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo keytool -list -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit
```

- 2. **Verify hostname resolution:**

```
# Test hostname resolution between nodes
ping hostname
host hostname
nslookup hostname
```

- 3. **Check certificate trust chain:**

```
openssl verify -CAfile /var/vcap/store/caspida/certs/my-root-ca.crt.pem /var/vcap/store/caspida/certs/my-server.crt.pem
```

Performance Issues

High CPU/Memory Usage

If the system experiences performance issues:

- 1. **Monitor system resources:**

```
top
htop
free -m
vmstat 1 10
```

- 2. **Monitor container resources:**

```
docker stats
docker ps
```

- 3. **Check for resource-intensive processes:**

```
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | head -20
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head -20
```

- 4. **Verify disk I/O performance:**

```
iostat -x 1 10
iotop
```

- 5. **Optimize data ingestion if necessary:**

```
# Edit /etc/caspida/local/conf/uba-site.properties
# Increase batch interval to reduce load
splunk.live.micro.batching.interval.seconds=120

# Synchronize changes
/opt/caspida/bin/Caspida sync-cluster /etc/caspida/local/conf
/opt/caspida/bin/Caspida restart
```

Disk Space Issues

If running out of disk space:

1. Check disk usage:

```
df -h
du -sh /var/vcap/* | sort -h
du -sh /var/vcap/store/* | sort -h
```

2. Clean up temporary files:

```
sudo rm -rf /tmp/*
sudo yum clean all
```

3. Clean up old log files:

```
# Clean old UBA logs (be careful with this)
find /var/vcap/sys/log -name "*.log.*" -mtime +7 -delete
```

4. Clean up old Docker containers and images:

```
docker container prune -f
docker image prune -f
docker system prune -f
```

Slow Web Interface

If the UBA web interface is slow:

1. Check UI service status:

```
sudo systemctl status caspida-ui
journalctl -u caspida-ui -f
```

2. Restart UI service:

```
sudo systemctl restart caspida-ui
```

3. Check browser compatibility:

- Use supported browsers (Chrome, Firefox, Safari)
- Clear browser cache and cookies
- Disable browser extensions

UBA Monitor API Errors

If you encounter errors like “Error from /uba/monitor - Invalid Json response: Error in getting the response - Parameters: {“queryStatus”:true,“queryDataQualityStatus”:true}“:

This error typically occurs when accessing the System > Health Monitor dashboard or when UBA’s internal monitoring API cannot retrieve system status information.¹

Error Analysis: The specific error pattern “Invalid Json response: Error in getting the response” indicates that the Job Manager’s REST API is either: - Returning malformed JSON due to an internal exception - Timing out during status queries - Encountering SSL/TLS handshake failures with internal services - Experiencing database connectivity issues during status aggregation

Quick Diagnosis Commands:

```

# First, verify UBA services are running
echo "Checking UBA service status:"

# Check if systemd services exist
if systemctl list-units --type=service | grep -q caspida; then
    echo "Found systemd UBA services:"
    systemctl list-units --type=service | grep caspida
    echo "Service status:"
    for service in $(systemctl list-units --type=service | grep caspida | awk '{print $1}'); do
        systemctl is-active "$service" || echo "$service is not active"
    done
else
    echo "Using Caspida script management - checking via ps:"
    ps aux | grep -E "(caspida|java.*uba)" | grep -v grep || echo "No UBA processes found"
fi

# Check if Job Manager API is responding at all
curl -k -X GET https://localhost:9002/health 2>&1 | head -5

# Check for immediate certificate/SSL issues
openssl s_client -connect localhost:9002 -verify_return_error </dev/null 2>&1 | grep -E "(Verify|error|failed)"

# Check for JSON parsing errors in Job Manager logs
# First verify log locations exist
if [[ -d "/var/log/caspida" ]]; then
    echo "Checking /var/log/caspida for JSON parsing errors:"
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -E "(JSON|json|parse|Invalid.*response)" 2>/dev/null | tail -10
elif [[ -d "/var/vcap/sys/log" ]]; then
    echo "Checking /var/vcap/sys/log for JSON parsing errors:"
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -E "(JSON|json|parse|Invalid.*response)" 2>/dev/null | tail -10
else
    echo "No standard UBA log directories found. Check actual installation paths."
fi

```

References: 1. Based on UBA deployment experience and analysis of common monitoring API failure patterns

Common Causes and Solutions:

1. Job Manager Service Issues:²

```

# Check Job Manager service status
sudo systemctl status caspida-jobmanager

# Check Job Manager logs for errors
echo "Checking for Job Manager service errors:"
if [[ -d "/var/log/caspida" ]]; then
    find /var/log/caspida -name "*jobmanager*" -type f 2>/dev/null | xargs tail -20 2>/dev/null
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -i "ERROR" 2>/dev/null | tail -10
elif [[ -d "/var/vcap/sys/log" ]]; then
    find /var/vcap/sys/log -name "*jobmanager*" -type f 2>/dev/null | xargs tail -20 2>/dev/null
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -i "ERROR" 2>/dev/null | tail -10
else
    echo "Please check actual UBA log locations on your system"
fi

```

2. Database Connectivity Problems:³

```

# Test PostgreSQL connection
sudo -u postgres psql -c "SELECT version();"

# Check database service status
sudo systemctl status postgresql

# Verify UBA can connect to database
sudo -u caspida psql -d caspidadb -c "SELECT count(*) FROM information_schema.tables;"

```

3. REST API Service Problems:⁴

```

# Check if Job Manager REST API is responding
curl -k -X GET https://localhost:9002/health

# Test with authentication token
TOKEN=$(grep 'jobmanager.restServer.auth.user.token' /opt/caspida/conf/uba-default.properties | cut -d'=' -f2)
curl -k -H "Authorization: Bearer $TOKEN" https://localhost:9002/health

```

4. Memory/Resource Constraints:⁵

```

# Check system resources
free -m
df -h

# Check Java heap usage for Job Manager
ps aux | grep jobmanager
jstat -gc <jobmanager_pid>

```

5. Configuration File Corruption:⁶

```
# Verify uba-site.properties syntax
grep -v "^#" /etc/caspida/local/conf/uba-site.properties | grep "="
# Check for configuration errors in logs
echo "Checking for configuration errors:"
if [[ -d "/var/log/caspida" ]]; then
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -i "configuration\|config" 2>/dev/null | tail -10
elif [[ -d "/var/vcap/sys/log" ]]; then
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -i "configuration\|config" 2>/dev/null | tail -10
else
    echo "Please check actual UBA log locations on your system"
fi
```

6. Java Keystore/Truststore Issues:¹³

```
# Check if Java truststore is accessible
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo keytool -list -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit | head -10

# Verify UBA keystore
sudo keytool -list -keystore /etc/caspida/conf/keystore/uba-keystore -storepass password
# Check for SSL handshake errors in Job Manager logs
echo "Checking for SSL/certificate errors:"
if [[ -d "/var/log/caspida" ]]; then
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -i "ssl\|certificate\|handshake\|truststore\|keystore" 2>/dev/null | tail -10
elif [[ -d "/var/vcap/sys/log" ]]; then
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -i "ssl\|certificate\|handshake\|truststore\|keystore" 2>/dev/null | tail -10
else
    echo "Please check actual UBA log locations on your system"
fi

# Test internal SSL connectivity
openssl s_client -connect localhost:9002 -verify_return_error
```

Resolution Steps:

1. Restart monitoring services:⁷

```
# Check if UBA is managed as systemd services or via Caspida scripts
echo "Checking UBA service management method:"
systemctl list-units | grep caspida || echo "No systemd caspida services found"

# Method 1: If UBA services are managed via systemd (newer installations)
if systemctl list-units | grep -q caspida; then
    echo "Using systemd service management:"
    # Check available caspida services
    systemctl list-units --type=service | grep caspida

    # Restart available UBA services
    for service in $(systemctl list-units --type=service | grep caspida | awk '{print $1}'); do
        echo "Restarting $service"
        sudo systemctl restart "$service"
    done
fi

# Method 2: If UBA is managed via Caspida scripts (standard method)
else
    echo "Using Caspida script service management:"
    # Stop all UBA services
    /opt/caspida/bin/Caspida stop-all

    # Wait for clean shutdown
    sleep 60

    # Start all UBA services
    /opt/caspida/bin/Caspida start-all
fi

# Wait for services to fully initialize
echo "Waiting for services to initialize..."
sleep 180
```

2. Clear temporary data:⁸

```
# Clear temporary monitoring cache
sudo rm -rf /tmp/caspida-monitor-*
sudo rm -rf /var/vcap/sys/tmp/monitor-*

# Clear browser cache and retry accessing Health Monitor
```

3. Verify service communication:⁹

```
# Test internal service connectivity
telnet localhost 9002
nc -zv localhost 9002

# Check firewall rules
sudo firewall-cmd --list-ports
sudo iptables -L | grep 9002
```

4. Database maintenance (if database issues suspected):¹⁰


```
# Restart PostgreSQL service
sudo systemctl restart postgresql

# Run database maintenance
sudo -u postgres psql -d caspidadb -c "VACUUM ANALYZE;"
```

5. If problem persists, collect diagnostic information:¹¹

```
# Capture current system state
/opt/caspida/bin/utils/uba_health_check.sh > /tmp/health_check.log

# Generate diagnostic bundle from UI: System > Download Diagnostics
# Collect relevant logs (adjust paths based on actual installation)
echo "Collecting UBA logs for troubleshooting:"
timestamp=$(date +%Y%m%d_%H%M%S)

# Create log collection directory
mkdir -p /tmp/uba_logs_$timestamp

# Collect logs from standard locations
if [[ -d "/var/log/caspida" ]]; then
    echo "Collecting logs from /var/log/caspida"
    cp -r /var/log/caspida/* /tmp/uba_logs_$timestamp/ 2>/dev/null || true
fi

if [[ -d "/var/vcap/sys/log" ]]; then
    echo "Collecting logs from /var/vcap/sys/log"
    mkdir -p /tmp/uba_logs_$timestamp/vcap_logs
    cp -r /var/vcap/sys/log/* /tmp/uba_logs_$timestamp/vcap_logs/ 2>/dev/null || true
fi

# Create archive
tar -czf /tmp/monitor_logs_$timestamp.tar.gz -C /tmp uba_logs_$timestamp/
echo "Log archive created: /tmp/monitor_logs_$timestamp.tar.gz"
```

6. Certificate-related troubleshooting (if SSL errors detected):¹⁴

```

# Check for common certificate-related error patterns in logs
# First, verify log directories exist and check available logs
echo "Checking available UBA log directories:"
ls -la /var/log/caspida/ 2>/dev/null || echo "Standard log directory not found"
ls -la /var/vcap/sys/log/ 2>/dev/null || echo "vcap log directory not found"

# Check main UBA logs for certificate errors (adjust path based on actual installation)
if [[ -d "/var/log/caspida" ]]; then
    echo "Checking /var/log/caspida for certificate errors:"
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -l -E "(certificate|keystore|truststore|SSL|TLS|handshake|PKIX|unable to
valid certification path)" 2>/dev/null | head -5
    echo "Recent certificate-related errors:"
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -E "(certificate|keystore|truststore|SSL|TLS|handshake|PKIX|unable to fi
valid certification path)" 2>/dev/null | tail -10
fi

# Check vcap logs if they exist
if [[ -d "/var/vcap/sys/log" ]]; then
    echo "Checking /var/vcap/sys/log for certificate errors:"
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -E "(certificate|keystore|truststore|SSL|TLS|handshake|PKIX|unable to f
valid certification path)" 2>/dev/null | tail -10
fi

# Validate certificate chain for Job Manager
openssl verify -CAfile /var/vcap/store/caspida/certs/my-root-ca.crt.pem \
/var/vcap/store/caspida/certs/my-server.crt.pem

# Check certificate expiration (alerts if expiring within 30 days)
openssl x509 -in /var/vcap/store/caspida/certs/my-server.crt.pem -noout -dates
openssl x509 -in /var/vcap/store/caspida/certs/my-server.crt.pem -noout -checkend 2592000

# Verify all UBA keystores
echo "Checking Job Manager keystore:"
openssl pkcs12 -info -in /etc/caspida/conf/keystore/keystore.jm -noout 2>/dev/null || \
    echo "ERROR: Job Manager keystore not accessible or corrupted"

echo "Checking main UBA keystore:"
sudo keytool -list -keystore /etc/caspida/conf/keystore/uba-keystore -storepass password 2>/dev/null || \
    echo "ERROR: UBA keystore not accessible or corrupted"

# Test SSL connectivity to all critical UBA services
echo "Testing Job Manager SSL:"
timeout 5 openssl s_client -connect localhost:9002 -verify_return_error 2>&1 | \
    grep -E "(Verify return code|SSL handshake|certificate verify)"

echo "Testing UI SSL (if HTTPS enabled):"
timeout 5 openssl s_client -connect localhost:443 -verify_return_error 2>&1 | \
    grep -E "(Verify return code|SSL handshake|certificate verify)"

# Check Java truststore integrity
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo keytool -list -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit 2>/dev/null | \
    grep -i "uba\|caspida" || echo "No UBA certificates found in Java truststore"

# Advanced certificate chain validation
echo "Performing complete certificate chain validation:"
for cert in /var/vcap/store/caspida/certs/*.crt.pem; do
    echo "Checking: $cert"
    openssl x509 -in "$cert" -noout -subject -issuer -dates -fingerprint
done

# Re-import certificates if corrupted or missing
echo "Re-importing UBA certificates to Java truststore:"
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo keytool -delete -alias "uba-ca" -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit 2>/dev/null || true
sudo keytool -delete -alias "uba-server" -keystore $JAVA_HOME/lib/security/cacerts -storepass changeit 2>/dev/null || true

# Import CA certificate
sudo keytool -import -trustcacerts -alias "uba-ca" -keystore $JAVA_HOME/lib/security/cacerts \
    -storepass changeit -file /var/vcap/store/caspida/certs/my-root-ca.crt.pem -noprompt

# Import server certificate if needed
sudo keytool -import -trustcacerts -alias "uba-server" -keystore $JAVA_HOME/lib/security/cacerts \
    -storepass changeit -file /var/vcap/store/caspida/certs/my-server.crt.pem -noprompt

# Restart services after certificate fixes
echo "Restarting UBA services after certificate remediation:"
/opt/caspida/bin/Caspida stop-all
sleep 30
/opt/caspida/bin/Caspida start-all

# Verify fix by testing monitor API
sleep 120 # Allow services to fully start
curl -k -X GET https://localhost:9002/health && echo "Job Manager API responding"

```

7. Advanced certificate debugging for specific error patterns:¹⁵

```

# For "PKIX path building failed" errors
echo "Diagnosing PKIX path building failures:"
openssl verify -verbose -CAfile /var/vcap/store/caspida/certs/my-root-ca.crt.pem \
/var/vcap/store/caspida/certs/my-server.crt.pem

# For "unable to find valid certification path" errors
echo "Checking Java certificate path validation:"
. /opt/caspida/bin/CaspidaCommonEnv.sh
sudo java -Djavax.net.debug=ssl:handshake:verbose \
-Djavax.net.ssl.trustStore=$JAVA_HOME/lib/security/cacerts \
-Djavax.net.ssl.trustStorePassword=changeit \
-cp /opt/caspida/lib/* com.sun.net.ssl.internal.www.protocol.https.HttpsURLConnection0ldImpl \
https://localhost:9002/health 2>&1 | head -50

# For certificate hostname validation errors
echo "Checking certificate Subject Alternative Names:"
openssl x509 -in /var/vcap/store/caspida/certs/my-server.crt.pem -noout -text | \
grep -A5 "Subject Alternative Name"
# Check if certificate matches current hostname
CURRENT_HOSTNAME=$(hostname -f)
echo "Current hostname: $CURRENT_HOSTNAME"
openssl x509 -in /var/vcap/store/caspida/certs/my-server.crt.pem -noout -text | \
grep -E "(CN=|DNS:|IP:)" | grep -i "$CURRENT_HOSTNAME" || \
echo "WARNING: Certificate may not match current hostname"

```

8. Post-certificate installation validation:¹⁶

```

# Complete validation after certificate changes
echo "Performing post-certificate installation validation:"

# Wait for services to stabilize
sleep 60

# Test monitor API specifically
echo "Testing UBA monitor API endpoint:"
curl -k -X GET "https://localhost:9002/uba/monitor?queryStatus=true&queryDataQualityStatus=true" \
-H "Content-Type: application/json" 2>&1 | head -10
# Check for certificate-related startup errors
echo "Checking for certificate errors in service startup:"
if [[ -d "/var/log/caspida" ]]; then
    find /var/log/caspida -name "*.log" -type f 2>/dev/null | xargs grep -E "(certificate|SSL|TLS|keystore|truststore)" 2>/dev/null | \
    grep -E "(ERROR|WARN|Exception)" | tail -10
elif [[ -d "/var/vcap/sys/log" ]]; then
    find /var/vcap/sys/log -name "*.log" -type f 2>/dev/null | xargs grep -E "(certificate|SSL|TLS|keystore|truststore)" 2>/dev/null | \
    grep -E "(ERROR|WARN|Exception)" | tail -10
else
    echo "Please check actual UBA log locations on your system"
fi

# Verify all UBA services can communicate
echo "Testing internal service communication:"
/opt/caspida/bin/Utils/test_service_connectivity.sh 2>/dev/null || \
    echo "Service connectivity test not available - manual verification required"

# Final health check
echo "Running comprehensive health check:"
/opt/caspida/bin/Utils/uba_health_check.sh | grep -E "(PASS|FAIL|ERROR)"

```

Prevention:¹² - Monitor system monitor resources regularly - Set up automated health checks - Ensure adequate disk space for temporary files - Keep PostgreSQL database properly maintained - Monitor certificate expiration dates and renew before expiry - Validate certificate chain integrity during maintenance windows - Backup certificate files before system upgrades

Network Issues

Service Communication Issues

If services cannot communicate with each other:

1. Check firewall settings:

```

# Check firewall status
sudo firewall-cmd --list-all

# Check specific ports
sudo firewall-cmd --list-ports

```

2. Test network connectivity:

```

# Test connectivity between nodes
telnet <other-node-ip> 9002
nc -zv <other-node-ip> 9002

```

3. Verify DNS resolution:

```

host <hostname>
nslookup <hostname>
dig <hostname>

```

RHEL 8.10 Network Configuration Issues

1. Check NetworkManager configuration:

```
sudo systemctl status NetworkManager
nmcli device status
nmcli connection show
```

2. Verify static IP configuration:

```
ip addr show
ip route show
```

Emergency Recovery Procedures

Reset Admin Password

If admin password is lost or not working:

1. Method 1 - Password reset script:

```
# Stop UI service
sudo systemctl stop caspida-ui

# Reset password
/opt/caspida/bin/admin_password_reset.sh

# Start UI service
sudo systemctl start caspida-ui
```

2. Method 2 - Database reset (if method 1 fails):

```
# Connect to database
sudo -u postgres psql -d caspidadb

# Update password to temporary value (password: admin123)
UPDATE users SET password = '$2a$10$Caspida123' WHERE username = 'admin';
\q

# Restart UI service
sudo systemctl restart caspida-ui
```

3. Login with temporary password and change immediately

Complete Service Reset

If UBA is completely unresponsive:

1. Stop all services:

```
/opt/caspida/bin/Caspida stop-all
```

2. Check and clean up processes:

```
# Check for hanging processes
ps aux | grep caspida
ps aux | grep kafka
ps aux | grep zookeeper

# Kill hanging processes if necessary
sudo pkill -f caspida
```

3. Clean temporary files:

```
sudo rm -rf /tmp/hisperfdata_*
sudo rm -rf /var/vcap/sys/tmp/*
```

4. Start all services:

```
/opt/caspida/bin/Caspida start-all
```

System Recovery After Reboot

If UBA doesn't start properly after system reboot:

1. Check mount points:

```
mount | grep vcap
df -h | grep vcap
```

2. Remount if necessary:

```
sudo mount -a
```

3. Start services in order:

```
# Start platform services first
sudo systemctl start postgresql
sudo systemctl start zookeeper-server
sudo systemctl start kafka-server

# Wait a few minutes, then start UBA services
/opt/caspida/bin/Caspida start
```

Getting Help

Collecting Diagnostic Information

Before contacting support, collect the following information:

1. Generate diagnostic bundle:

```
# Navigate to System > Download Diagnostics in UBA UI
# Select all components and download
```

2. System information:

```
# UBA version
cat /opt/caspida/conf/version.properties

# OS information
cat /etc/redhat-release
uname -a

# Hardware information
free -m
df -h
lscpu
```

3. Recent log entries:

```
# Last 100 lines of key logs
tail -100 /var/vcap/sys/log/caspida-ui/caspida-ui.stderr.log
tail -100 /var/vcap/sys/log/caspida-jobmanager/caspida-jobmanager.stderr.log
tail -100 /var/log/caspida/caspida.out
```

Support Information

- **Version:** Splunk UBA 5.4.2 on RHEL 8.10
- **Document Version:** 1.0
- **Last Updated:** June 2025

For technical support, provide: - UBA version and build number - RHEL 8.10 version and kernel information - Deployment topology (single/distributed) - Description of issue with relevant log entries - Output of health check script - Diagnostic bundle (if possible)

References

UBA Monitor API Errors Section

1. UBA Monitor API error analysis based on field experience and common monitoring dashboard failures
2. Splunk UBA Job Manager service troubleshooting - derived from UBA service architecture documentation
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Distributed>
3. PostgreSQL connectivity testing for UBA - based on UBA database configuration requirements
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Prerequisites>
4. UBA REST API testing procedures - based on UBA Job Manager API documentation and internal service communication patterns
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/GetData/Monitorhealth>
5. System resource monitoring for UBA - derived from UBA performance optimization guidelines
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Systemrequirements>
6. UBA configuration file validation - based on UBA site properties documentation and common configuration errors
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Configure>
7. UBA service restart procedures - based on UBA service management best practices
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/StartUBA>
8. Temporary file cleanup for UBA - based on UBA system maintenance procedures

- <https://docs.splunk.com/Documentation/UBA/5.4.2/Admin/Maintenance>
 - 9. Service communication verification - derived from UBA network troubleshooting procedures
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Firewallports>
 - 10. PostgreSQL maintenance for UBA - based on UBA database maintenance requirements
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Admin/Maintenance>
 - 11. UBA diagnostic data collection - based on Splunk UBA troubleshooting and support procedures
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Admin/Troubleshooting>
 - 12. UBA preventive maintenance - derived from UBA operational best practices and system hardening guidelines
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Admin/Maintenance>
 - 13. Java Keystore/Truststore Issues - based on UBA SSL/TLS configuration and certificate management procedures
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Configure>
 - 14. Certificate troubleshooting and remediation - derived from UBA certificate management best practices and SSL/TLS debugging procedures
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Certificates>
 - 15. Advanced certificate debugging and PKIX path validation - based on Java SSL debugging techniques and UBA certificate chain validation procedures
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Certificates>
 - 16. Post-certificate installation validation procedures - derived from UBA certificate deployment best practices and service integration testing
 - <https://docs.splunk.com/Documentation/UBA/5.4.2/Install/Certificates>
-

Note: This troubleshooting guide should be used in conjunction with the main Splunk UBA Deployment Guide and Deployment Checklist documents.